# Hieu Vo

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## Education

**Villanova University** Expected May 2028

*B.S. in Computer Engineering, Minor in Computer Science and Cybersecurity* *GPA: 3.6/4.0*

- **Coursework:** Analysis of Algorithm, Database Systems, Computer Networks, Web and Mobile design, C++ Algorithms and Data Structures, Discrete Structures, Digital Systems, Signal Processing
- **Honors/Awards:** Dean's List (All terms), RevolutionUC Hackathon Winner Best Use of MongoDB Atlas

## Experience

**BriteAI** May 2026 – Present

*Software Engineer Intern* *Mountain View, CA*

- Built a notification service using **Java**, **Spring Boot**, **CockroachDB**, and **WebSocket/STOMP**, supporting 3 notification workflows across 2 delivery channels: real-time in-app updates and email
- Reduced **RabbitMQ** payload size by **93%** (from **6 KB** to **400 bytes**/event) by sending events containing only IDs and metadata, while cutting preference lookups from **1,001** database queries to **2** queries for 1,000 subscribers
- Added an in-memory cache for unread notification counts, reducing count lookup time from **20–50 ms** to under **5 ms**
- Automated deployment of a **Spring Boot** service to **Google Compute Engine**, reducing deployment time by **75%**, while securing internal APIs with **HMAC-SHA256** and adding health checks for database pools and background workers

**Media Landscape (MeLa) Lab** Oct. 2025 – May 2026

*Software Engineer Intern* *Villanova, PA*

- Built a **Django REST** and **PostgreSQL** analytics platform over **40M+** news articles, enabling researchers to compare coverage, temporal, linguistic, and stylistic media bias across **500+** outlets; optimized database indexes to reduce query latency by **40%**(from **2.5** to **1.5 seconds**)
- Architected a NLP pipeline using **Kafka**, **Celery**, and **Redis**, dividing workloads across parallel workers to increase processing throughput by **4x**(from 15K to 60K+ articles/day), reducing batch processing time by **80%**(from 5 to 1 hour)
- Engineered **10+** automated **Python** data pipelines using **spaCy** and **NLTK** to process **60k+** articles daily from **500+** outlets, enabling high volume feature extraction and analytics via **10** API endpoints
- Built a **BERTopic** pipeline using **Sentence Transformers**, **UMAP**, and **HDBSCAN** to identify **17+** major news topics from **50K** articles and enable scalable topic assignment across a **40M+** article corpus

**L3Harris Technologies** Jun. 2025 – Jul. 2025

*Software Engineer Intern* *Villanova, PA*

- Engineered a high-throughput image-processing pipeline using **Python** and **OpenCV** to convert **4K+** wind tunnel videos into **12K+** paired training images for an AI-based airfoil screening system
- Improved compute efficiency through LoRA fine-tuning, adaptive batch sizing, and GPU memory footprint optimization, reducing training time by **80%** (from 10 to 2 hours) and infrastructure cost by **70%**
- Launched an AI-driven aerodynamic screening service using **FastAPI, MySQL, React**, and **Docker CI/CD**, providing scalable on demand airflow predictions and enabling **98%** faster design iteration

## Projects

**ClinSync – RevolutionUC Winner** | *FastAPI, MongoDB, Gemini API, LangChain/LangGraph, ChromaDB, Tesseract.js*

- Architected a clinical trial recruitment platform using **FastAPI**, **MongoDB**, and **React.js**, orchestrating multi-agent AI workflows for automated eligibility screening and reducing patient–trial matching latency by **50%**
- Re-designed OCR processing to run client-side using **Tesseract.js**, decreasing backend compute utilization by **15%** and cutting server-side exposure of sensitive health documents by **99%**
- Built a structured retrieval and reasoning pipeline using **ChromaDB** and **LangChain/LangGraph** with Gemini-powered agents, improving trial-matching accuracy by **20%**

**FixBuddy** | *Node.js, Express, MongoDB, Gemini API, LangChain*

- Led a team of 2 to architect an AI-powered repair platform with modular backend services built on **Node.js**, **Express**, and **MongoDB**, reducing average API response time by **45%**
- Implemented a multi-agent architecture with **Gemini API** and **LangChain**, separating item diagnosis from repair instruction generation to improve reasoning consistency and reduce error rates by **70%**
- Optimized **RESTful API** design and database indexing strategies to support concurrent users, decreasing database query latency by **35%** and improving backend stability

## Technical Skills

**Languages**: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML, CSS, MATLAB
**Frameworks & Libraries**: Spring Boot, WebSocket/STOMP, React, Next.js, Node.js, FastAPI, Django REST, Express.js, Flask, Celery, HikariCP, PyTorch, Scikit-Learn, spaCy, NLTK, LangChain, LangGraph, OpenCV
**Databases**: CockroachDB, PostgreSQL, MongoDB, MySQL, Redis, RabbitMQ, Kafka, ChromaDB
**Developer Tools**: Google Compute Engine, Docker, Maven, Flyway, systemd, Git, GitHub, Postman, cron